Supplementary Material: Feynman-Diagram Notation of the LOKLOK Experimental Processes

Sherif Mohamed Abd Elmonem Rashwan
Ahram Canadian University, Faculty of Mass Communication

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Companion supplement to the master's thesis "The Social Media Radio Innovation and Its Impact on University Students' Ratings: An Exploratory Study."

Purpose and convention

The diagrams in this supplement express the LOKLOK experimental processes in the diagrammatic notation of particle physics. They are pedagogical supplements: they borrow the visual grammar of Feynman diagrams to make mechanism dependencies, interaction orders, and measurement points explicit. Nothing in them is a claim about physics; the Hamiltonian analogy in the thesis carries the same caveat.

Table 1: Line and vertex conventions

Symbol	Meaning in the LOKLOK reading
Solid arrow line (fermion)	Agent line: student, listener, broadcaster, viewpoint, or content; the arrow gives the directio
Wiggly line (photon)	Audio signal, broadcast, reflection prompt, cooling radiation, or a measurement response
Curly line (gluon)	Platform mechanism exchange: diversity constraint, LOKLOK filters, gate self-interaction, e
Dashed line (scalar)	Misinformation before and corrected content after moderation
Filled circle (blob)	Algorithmic filter or update: MORE, ERIC, BRS evidence update, Zewail gate, GRM-IRT c
Crossed vertex	Randomization point or measurement point separating latent from observed

The ten situations

D1. Listening as absorption

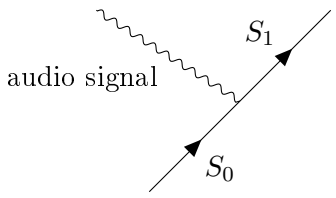


Figure 1: D1. Listening as absorption

A broadcaster emits an audio signal (photon). The listener line (fermion) absorbs it at a vertex and continues with an updated judgment state S0 to S1. This is the elementary process every other diagram composes.

D2. EST-P1: structural credibility through the BRS anchor

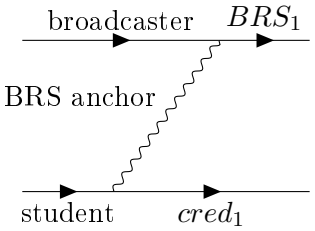


Figure 2: D2. EST-P1: structural credibility through the BRS anchor

A student and a broadcaster exchange a BRS anchor (photon) at a scattering vertex. After the exchange the student carries credibility cred1 and the broadcaster carries reputation BRS1: architecture enters as the exchanged field, not as an external parameter.

D3. EST-P2: the diversity constraint in MORE

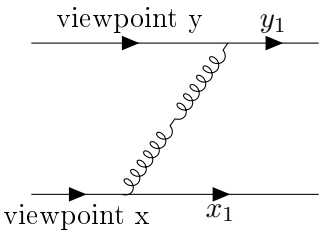


Figure 3: D3. EST-P2: the diversity constraint in MORE

Two viewpoints exchange a diversity constraint through the MORE filter (blob), drawn as a gluon exchange. The blob makes the constraint local to the platform; the viewpoints emerge correlated rather than independent.

D4. The Zewail-inspired attention gate

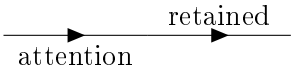


Figure 4: D4. The Zewail-inspired attention gate

An attention line passes through the gate (blob) and emerges retained. The gluon loop on the blob is the gate self-interaction, parameterized by the dephasing time tau: the loop stands for the time-resolved retention that a static filter does not have.

D5. Algorithmic cooling of echo

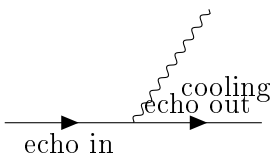


Figure 5: D5. Algorithmic cooling of echo

An echo line radiates a cooling photon at a vertex, like bremsstrahlung. The radiated energy is removed from the echo state, which is the diagrammatic form of the cooling step.

D6. The main experiment: G0 to G1

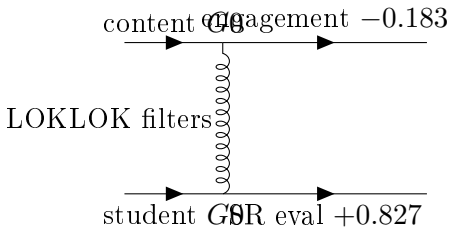


Figure 6: D6. The main experiment: G0 to G1

A student line and a content line enter from G0 and exchange the LOKLOK filter field (gluon). The outgoing states carry the headline observables: SR eval +0.827 and engagement -0.183. The diagram makes the interaction order explicit: the platform acts between the student and the content, never on them separately.

D7. The BRS evidence update

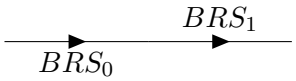


Figure 7: D7. The BRS evidence update

A reputation line passes through an update blob. The gluon loop injects evidence at the blob: each loop order corresponds to one round of evidence-based revision, the diagrammatic form of the reputation contraction.

D8. ERIC moderation as a correction vertex

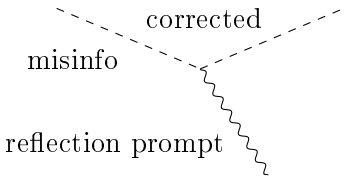


Figure 8: D8. ERIC moderation as a correction vertex

A misinformation line (scalar, dashed) enters the ERIC blob and leaves corrected. The radiated photon is the reflection prompt sent back to the user: moderation is drawn as an interaction, not as deletion.

D9. Randomized assignment as a branching vertex

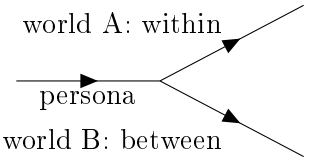


Figure 9: D9. Randomized assignment as a branching vertex

A persona line splits at a crossed vertex into two worlds: within-persona assignment (A) and between-agent assignment (B). The crossing marks the randomization point, the only place where the two designs differ.

D10. GRM-IRT measurement

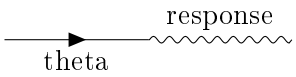


Figure 10: D10. GRM-IRT measurement

A latent trait line (theta) reaches a crossed measurement vertex and emits a response photon that the detector blob (GRM-IRT scoring of the NMLS and MCS batteries) absorbs. The cross separates the unobservable trait from the observed rating.